

Aerodynamic characterisation of *NACA 0012* and *NACA 2412* aerofoils at low Reynolds numbers

Luis Gustavo Gomes Maia[†]

[†]Instituto Tecnológico de Aeronáutica

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1 Introduction

In this work, an experimental study of two classical aerofoil sections, the symmetric *NACA0012* and the cambered *NACA 2412*, was conducted in a subsonic wind tunnel. Measurements were performed using an aerodynamic balance capable of resolving drag, lift, and pitching moment, together with a pressure transducer for flow characterization. The effects of Reynolds number and flap deflection were investigated in order to assess how camber and control-surface configuration influence the aerodynamic behavior, especially near stall conditions.

The collected data were used to determine lift and drag coefficients as functions of angle of attack, as well as to evaluate the validity of simplified aerodynamic models in the pre-stall regime. Particular

attention was given to the linear region of the lift curve, the maximum lift coefficient, and the drag polar representation. The results provide insight into the influence of camber and flap deflection on aerodynamic performance and illustrate the limitations of theoretical approximations when compared with experimental observations.

2 Methodology

The methodology, both theoretical and experimental, together with its post-processing, considered in this report is described below.

2.1 Theoretical

From the equilibrium of forces and moments, one can calculate the moment coefficient about any point from Eq. 1:

$$c_m = c_m^{r_0} - (X - X_{r_0})c_l \quad (1)$$

where c_m and c_l are the moment and lift coefficients, respectively, r_0 is a reference point — in the x or wind direction — and X is this normalised coordinate.

Considering the definition of the aerodynamic centre, according to which the moment coefficient does not change with lift, or $\frac{\partial c_m^{ca}}{\partial c_l} = 0$, one obtains:

$$X_{ca} = X_{r_0} + \frac{\partial c_m^{r_0}}{\partial c_l} \quad (2)$$

With respect to the lift curve, for small angles of attack it is linear; however, it reaches a maximum and decays due to drag effects. Thus, one may propose a variation as

$$c_l = c_l(\alpha) = a_0 + a_1\alpha + a_2e^{a_3\alpha} \quad (3)$$

where a_2 is certainly negative. This dependence should be valid for angles somewhat greater than the stall angle.

Regarding drag, one may also propose a variation with lift, parabolic in this case:

$$c_d = c_d(c_l) = c_{d,0} + k_1c_l + k_2c_l^2 \quad (4)$$

Other aspects of the aerofoil behaviour can be considered in light of thin-aerofoil theory [1]. According to this theory, the zero-lift angle of attack depends on the shape as

$$\alpha_0 = -\frac{2}{\pi} \int_0^\pi \frac{Z^{(s)}(\phi)}{1 - \cos\phi} d\phi \quad (5)$$

where $Z^{(s)} = Z^{(s)}(\phi)$ is the mean camber ordinate of the aerofoil.

Finally, both the lift slope and the (normalised) aerodynamic centre should be 2π and $\frac{1}{4}$, respectively, regardless of the aerofoil camber.

2.2 Experimental

The aerodynamic balance attached to the wind tunnel has three load cells, one of which is aligned with the wind direction and the other two perpendicular to it. There is a separation distance of b between their positions. Thus, one can calculate the lift, drag and pitching moment:

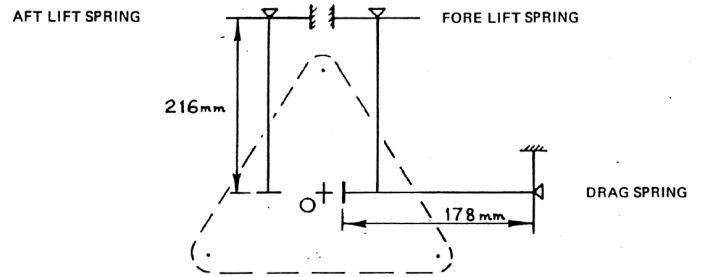
$$D = F_D \quad (6)$$

$$L = F_A + F_F \quad (7)$$

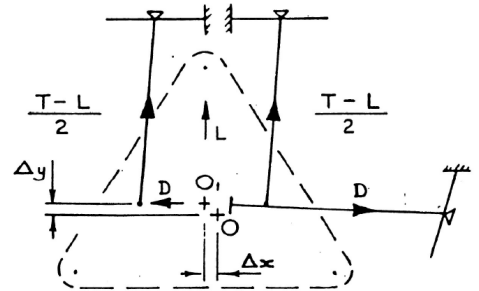
$$M = b(F_F - F_A) \quad (8)$$

where F_D , F_A and F_F are the measurements provided by the balance.

It should be noted that these relations are approximations, since any force applied to the scale induces displacements that affect all three cells. The correlation, however, is insignificant, as will be shown in the calibration procedure. This can be seen in Fig. 1b:



(a) Scheme of the balance and measurements.



(b) Scheme of the distorted balance.

Figure 1: Schemes of the balance [2].

The calibration of the scale was carried out by placing masses on a plate attached to it, increasing the load directly in the two directions of interest and then decreasing it. This procedure was per-

formed to evaluate the hysteresis of the balance. For drag, masses of 100 g were used until the maximum supported load of 1 kg was reached. On the other hand, for lift cells, which supports up to 4 kg, masses of 500 g and 1 kg were used. During the measurements, a vibrator attached to the tunnel was activated to settle the scale, increasing its precision and reducing the settling time.

The calibration of the pressure transducer was carried out by connecting it to a chamber whose pressure, increased and decreased, was measured using a Betz manometer. The latter was first filled with water to the top to avoid variations in water level during the measurements. Its pressure was set using a manual pump and the reference value was recorded manually. It is relevant to note that a large pressure variation occurred during the readings, which naturally increases the uncertainties of the reference measurements.

Regarding the measurements, the aerofoil sections were placed in the tunnel and attached to the balance. The wind speed was increased until the velocity required to reach the desired Reynolds numbers was attained. Two different Reynolds numbers were chosen for the aerofoils without flaps, and only one for the flap cases of the *NACA 2412*. They were $2.2 \cdot 10^5$ and $3.1 \cdot 10^5$. Finally, the angles of attack were increased from negative to positive values until stall occurred. Before changing the aerofoils and flow velocities, a null measurement of the balance cell outputs was taken with no wind.

2.3 Data handling

The first data to be processed were those from the calibration procedures. The raw voltage values from the three load cells, associated with the masses applied to the scale, were fitted by Weighted Least Squares to a straight line — as expected given its construction and the weights used. These functions were then used to convert the voltage signals measured during the experiments into masses (forces), including their prediction bounds. Measurement outliers were removed if they lay beyond a quantile with probability $1/(4n)$ from the predictions, where n is the number of

points in the sample. This is in accordance with Chauvenet’s criterion [3]. The same procedure was applied to the pressure transducer.

Once the tests with the aerofoils had been carried out according to the experimental methodology described, the data acquisition system provided, for each case, the sensor voltage values as a function of the angle of attack. Applying the calibration curves yielded the mass values measured by the load cells, D , A and F . The first measurement of each run — taken with the wind tunnel stopped — was subtracted from the others and then excluded from the dataset. Afterwards, the corresponding lift, drag and moment values were obtained by multiplying by gravitational field and applying Eqs. 7, 6 and 8.

The ambient air density and its kinematic viscosity were calculated from the measurements of pressure and temperature at the beginning of each run. Assuming atmospheric air to behave as a perfect gas, the ideal gas equation, Eq. 9, was used:

$$\rho = \frac{p}{RT} \quad (9)$$

where ρ is the density, p the pressure, T the temperature of the atmospheric air, and R its specific gas constant.

In order to calculate the viscosity of air under the experimental conditions, the Sutherland model was adopted, according to which this quantity varies with temperature as in Eq. 10 [4]:

$$\frac{\mu}{\mu_0} = \left(\frac{T}{T_0} \right)^{3/2} \frac{T_0 + S}{T + S}, \quad (10)$$

where μ_0 , T_0 and S are, respectively, the reference viscosity, reference temperature and Sutherland constant.

The dynamic pressure values were obtained from the transducer readings in $\text{mm}_{\text{H}_2\text{O}}$ for each measurement point. The corresponding velocities were found from

$$q_\infty = \frac{1}{2} \rho_\infty U_\infty^2 = \rho_{Ag} gh \quad (11)$$

The Reynolds number was also calculated. Finally, the lift, drag and moment coefficients were determined.

The coefficients a_0 to a_3 of the model presented in Eq. 3 were obtained by non-linear fitting of the data points. In the case of the drag model, Eq. 4, Weighted Least Squares was used.

With these coefficients, it was possible to calculate the zero-lift angle, the lift slope, the stall angle and the maximum c_l predicted by the model. The first was obtained by solving the implicit equation $0 = a_0 + a_1\alpha_0 + a_2e^{a_3\alpha_0}$. The second, for this non-linear lift model, is $c_{l_\alpha} = [dc_l/d\alpha]_{\alpha=0} = a_1 + a_2a_3$. The third follows from $[dc_l/d\alpha]_{\alpha=\alpha_e} = 0 \implies a_1 + a_2a_3e^{a_3\alpha_e} = 0 \therefore \alpha_e = 1/a_3 \log(-a_1/a_2a_3)$. Finally, the maximum lift coefficient is obtained directly from $c_l(\alpha_e) = a_0 + a_1\alpha_e + a_2e^{a_3\alpha_e}$.

For the calculation of the aerodynamic centre, X_{ca} , considering the linear region $\alpha \in [-5^\circ, 5^\circ]$, the derivative dc_m/dc_l was estimated as the slope between these quantities. Eq. 2 was then applied.

3 Results

The results of the experiments are presented below.

3.1 Experimental conditions

The experimental conditions measured and calculated for each test are presented in Table 1:

Table 1: Ambient conditions and air properties measured during the tests.

Aerofoil	Flap	U_∞	T (K)	p (10^4 Pa)	ρ (kg/m^3)	ν ($10^{-5} m^2/s$)
<i>NACA 0012</i>	0	25	$297,5 \pm 0,5$	$9,52 \pm 0,01$	$1,115 \pm 0,002$	$1,643 \pm 0,005$
		35	$297,8 \pm 0,5$	$9,52 \pm 0,01$	$1,114 \pm 0,002$	$1,647 \pm 0,005$
<i>NACA 2412</i>	0	25	$298,3 \pm 0,5$	$9,52 \pm 0,01$	$1,112 \pm 0,002$	$1,652 \pm 0,005$
		35	$298,5 \pm 0,5$	$9,52 \pm 0,01$	$1,111 \pm 0,002$	$1,654 \pm 0,005$
	15	35	$298,8 \pm 0,5$	$9,52 \pm 0,01$	$1,110 \pm 0,002$	$1,657 \pm 0,005$
	30	35	$299,0 \pm 0,5$	$9,52 \pm 0,01$	$1,109 \pm 0,002$	$1,659 \pm 0,005$

For the calculations according to the methodology presented, the following reference values were used:

Table 2: Reference values of temperature, viscosity and Sutherland constant used.

Quantity	Value
μ_0 ($10^{-5} kg/ms$)	1,79
T_0 (K)	288,16
S (K)	110
g (N/kg)	9,7685
ρ_{Ag} (kg/m^3)	997,8

3.2 Calibration

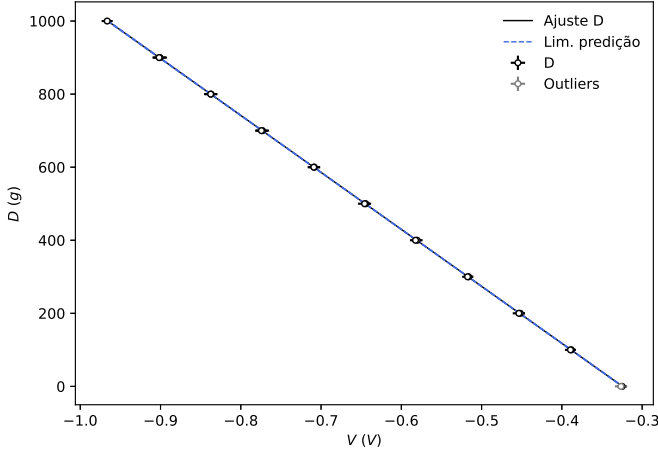
For the calibration of the balance and the transducer, linear relations between the physical quantities and the voltage levels were expected. The corresponding lines, according to $y = a_0 + a_1x$, have coefficients presented in Table 3:

Table 3: Linear fit coefficients with 95% confidence.

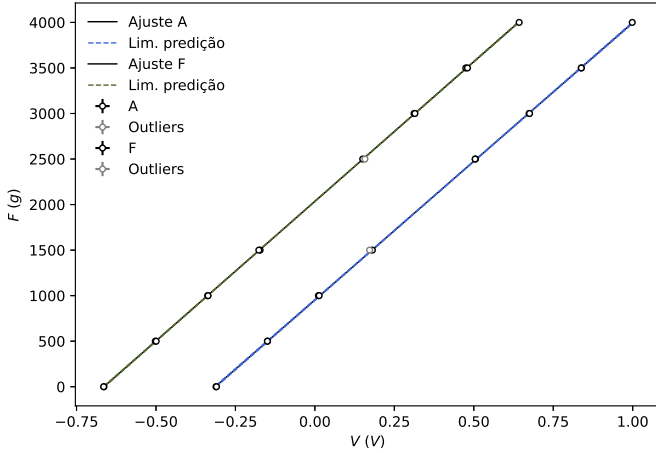
Sensor	a_0	a_1	r^2
p	$-0,32 \pm 0,03$	$-7,63 \pm 0,01$	1,000
D	$-506,0 \pm 0,8$	-1560 ± 1	1,000
A	953 ± 3	3044 ± 5	1,000
F	2035 ± 2	3066 ± 4	1,000

All sensors exhibited a correlation coefficient r^2 equal to 1.000 to the reported precision – up to fourth decimal –, indicating an essentially linear relationship.

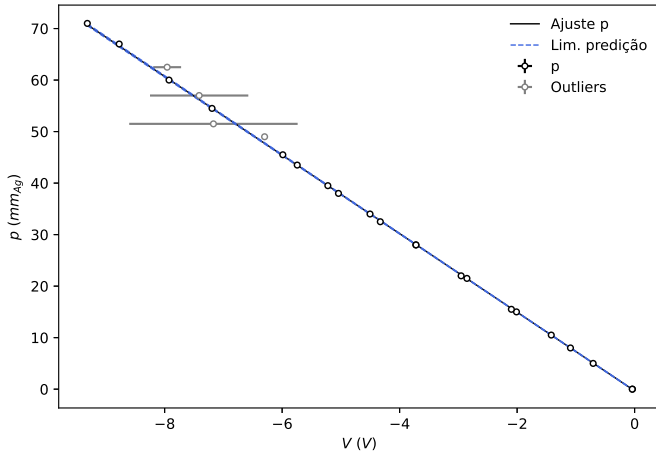
The measurements and fitted lines can be seen in Fig. 2:



(a) Load cell D of the balance.



(b) Load cells A and F of the balance.



(c) Pressure transducer.

Figure 2: Calibration of the balance load cells and pressure transducer.

A high degree of linearity of the data points can

be observed, as expected, especially for the load cells.

For the D cell, a uniform distribution of masses was chosen, whereas for the A and F cells a distribution denser at lower and higher masses was preferred, corresponding to lower and higher angles of attack, respectively. Another advantage of this choice is that the mean of the measured points is not displaced from the mean of the applied masses, which naturally increases precision in this measurement range.

Finally, for the transducer, some points deviating from the fitted line can be observed. This was significant due to the rapid pressure decay in the chamber, as explained previously. Their uncertainties were sufficiently large to justify concern. These outliers were removed according to Chauvenet's criterion.

3.3 Coefficients and aerodynamic characteristics

From the variation proposed in Eq. 3, the coefficients a_0 to a_3 were obtained, from which the zero-lift angle of attack α_0 , the lift slope c_{l_α} , and the normalised aerodynamic centre X_{ca} were calculated. The values of the former coefficients are shown in Tab. 4 and those of the latter in Tab. 5.

The dynamic pressure inside the tunnel was also measured using a Total Pitot tube. As the aerofoil angle of attack changed during the experiment, the blockage effect was not compensated by the motor, so that the wind velocity – and thus the Reynolds number – decreased. The Reynolds numbers presented in Tab. 5 correspond to zero angle of attack.

A deviation from the expected Reynolds number can be observed, especially for aerofoils with flap deflection, because this element causes greater blockage, which the tunnel motor could not compensate for even at zero angle of attack.

Table 4: Aerodynamic coefficients obtained by regression.

Aerofoil	Flap	v	a_0	a_1	a_2	a_4
<i>NACA 0012</i>	0	25	$0,004 \pm 0,04$	$7,3 \pm 0,3$	$-0,04 \pm 0,03$	12 ± 2
		35	$0,000 \pm 0,09$	$7 \pm 0,6$	$-0,06 \pm 0,07$	11 ± 4
<i>NACA 2412</i>	0	25	$0,20 \pm 0,05$	$7,2 \pm 0,6$	$-0,02 \pm 0,03$	15 ± 7
		35	$0,26 \pm 0,1$	$8,0 \pm 0,8$	$-0,07 \pm 0,1$	10 ± 5
	15	35	$0,88 \pm 0,06$	$7,2 \pm 0,7$	$-0,003 \pm 0,01$	28 ± 20
	30	35	$1,4 \pm 0,06$	$6,8 \pm 0,7$	$-0,005 \pm 0,02$	27 ± 20

Table 5: Experimental aerodynamic parameters.

Aerofoil	Flap	v	$Re (10^5)$	$\alpha_0 (^\circ)$	c_{l_α}	X_{ca}
<i>NACA 0012</i>	0	25	$2,605 \pm 0,009$	$0,0 \pm 0,3$	$6,8 \pm 0,5$	$0,254 \pm 0,003$
		35	$3,06 \pm 0,01$	$0,0 \pm 0,7$	7 ± 1	$0,247 \pm 0,003$
<i>NACA 2412</i>	0	25	$1,945 \pm 0,007$	$-1,6 \pm 0,4$	$7,0 \pm 0,7$	$0,17 \pm 0,05$
		35	$2,290 \pm 0,008$	$-1,9 \pm 0,9$	7 ± 1	$0,228 \pm 0,006$
	15	35	$2,265 \pm 0,008$	$-7,0 \pm 0,8$	$7,1 \pm 0,8$	$0,28 \pm 0,05$
	30	35	$2,308 \pm 0,008$	-12 ± 1	$6,7 \pm 0,9$	$0,17 \pm 0,05$

Regarding the zero-lift angle, it was found to be approximately zero for the *NACA 0012*, as expected – except for the uncertainty. A decrease of about 2° is observed due to the camber of the aerofoil. A much larger decrease is observed for the aerofoils with flaps. This decrease appears to be nearly linear with flap deflection, as there is a reduction of approximately 5° both from 0° to 15° and from 15° to 30° . This is consistent with the linearity predicted by thin-aerofoil theory, as shown in Eq. 5.

Finally, the aerodynamic centre of the *NACA 0012* was close to the quarter-chord, as expected, for both Reynolds numbers tested, with an accuracy better than $\pm 2\%$. The same was not true for the *NACA 2412*. Although the case with the smallest uncertainty was closest to the theoretical value of $1/4$, it still showed an inaccuracy of -9% . The others, with -32% and 12% , do not allow further conclusions.

These higher uncertainties arise from the attempt to fit a four-parameter model to a small dataset – of 16 points –, which, due to experimental deviations, did not strictly follow the expected behaviour. Factors contributing to this include ac-

commodation effects in the balance (mainly), tunnel vibrations, and changes in the airflow downstream of the test section, considering that the tunnel is open immediately after it. This type of uncertainty could not be quantified and is therefore not included in the calculated standard deviations.

3.4 Lift coefficient

The lift-curve graphs are presented in Fig. 3, together with the fitted curves of the model.

Mainly for the *NACA 0012*, one observes a highly linear behavior for small angles of attack, between -5° and 5° . After reaching the maximum at the stall limit, the lift starts to decay. This decay is delayed for the *NACA 2412* without flap, the lift being sustained for a longer interval after stall. The cases with flap exhibit both behaviors.

Another difference is that, for the 2412, even at small angles, there are some points that deviate from the initial straight line, which were common to both Reynolds number experiments without flap – e.g., for $\alpha = 6^\circ$ – and with flap – e.g., the points for $\alpha = -4^\circ, 0^\circ, 2^\circ$, and 4° .

As can also be noted, the camber of the 2412

aerofoil causes an increase in the maximum lift – around 1.4 compared to 1.0 for the 0012 – but, in this case, without a significant anticipation of stall, which occurs near 12° , as for the corresponding symmetric profile. In the cases with flap, the increase is much larger; however, there is a shift to approximately 10° and 8° for the 15° and 30° flap deflections, respectively. These displacements occur for all other points of the curves. The effect of camber on the lift–alpha curve can thus be interpreted as a translation toward the second quadrant.

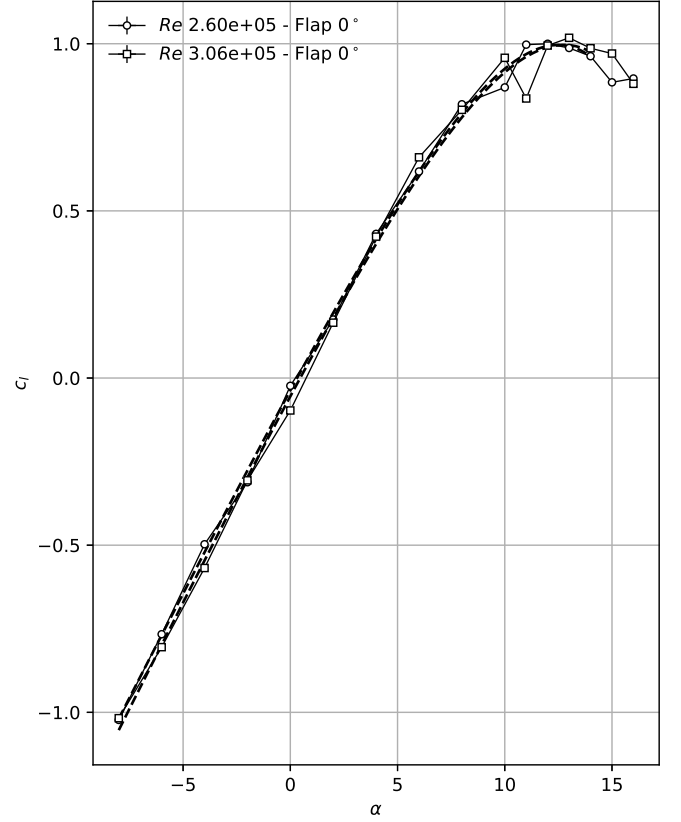
Nevertheless, given the general behavior observed, it is possible to conclude that the model proposed in Eq. 3 represents it with simplicity and sufficient accuracy, at least up to stall. This is also observed in the same graphs.

3.5 Drag coefficient

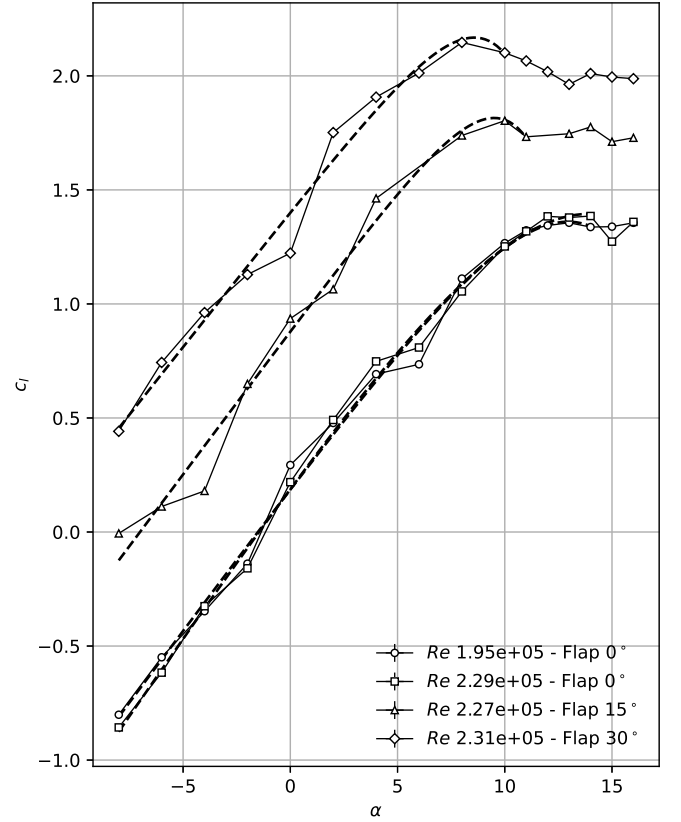
Regarding the drag coefficient, the points are even further from an expected smooth trend in comparison to lift, especially for the *NACA 2412*. Although this makes the analysis more difficult, it does not prevent one from observing a slight increase in drag up to the pre-stall region. To describe this behavior, a parabolic model is proposed, as in Eq. 4. It is noteworthy that it is expressed as a function of lift, not angle of attack. This would be irrelevant if the relationship between these two quantities were strictly linear. Note the decrease for negative α in the case of the 30° flap, where the lift is positive. This occurs because, locally along the cambered aerofoil, the surface angle is on average larger than that of the non-cambered profile. The higher pressure gradients thus increase the local friction drag, which also explains the earlier separation of the boundary layer and consequent. The generally larger values of c_d for the 2412 compared to the 0012 provide evidence of this influence.

With further increases in the angle of attack, the drag grows considerably after stall, as expected.

The graphs in Fig. 5 show the fitted parabolas to the points prior to stall. Visually, the fit is good for the *NACA 0012*, despite the deviations mentioned, but appears inconclusive for the 2412 in all its cases.

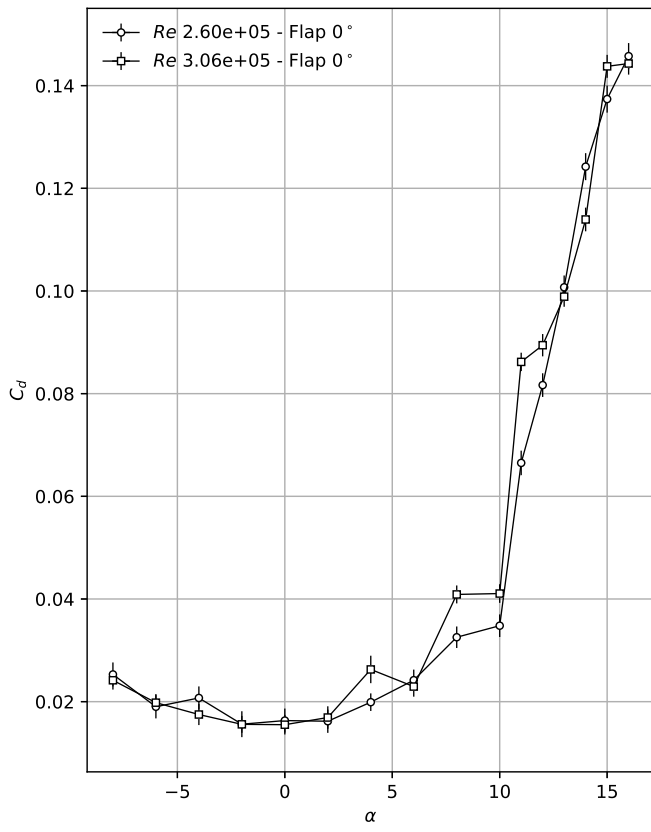


(a) For NACA 0012.

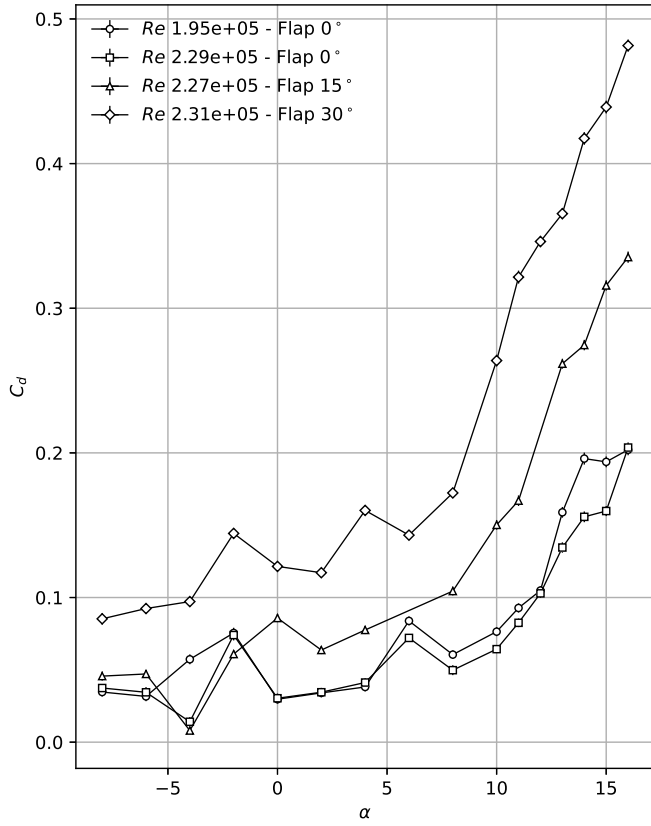


(b) For NACA 2412.

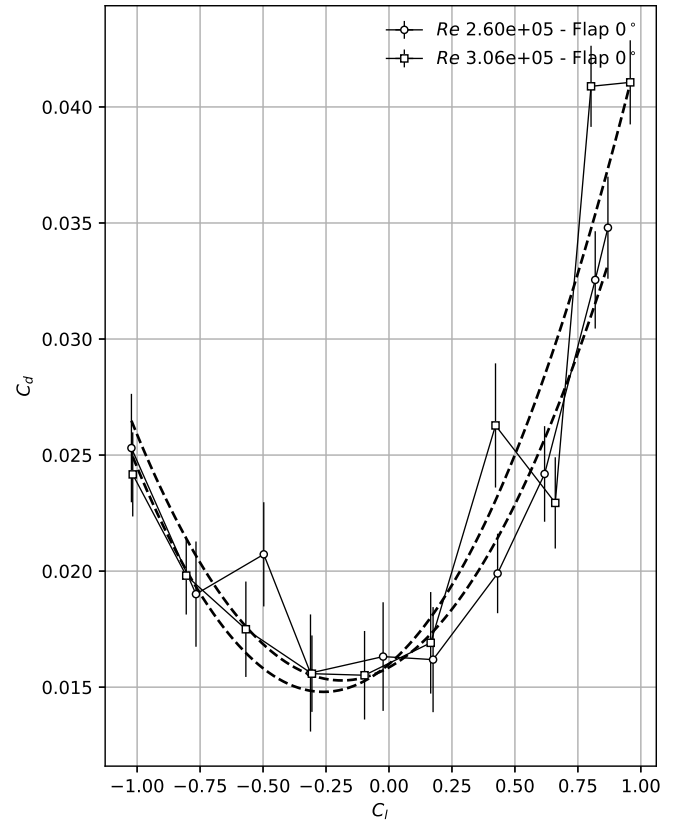
Figure 3: Lift coefficient c_l as a function of angle of attack α and exponential adjustment until stall.



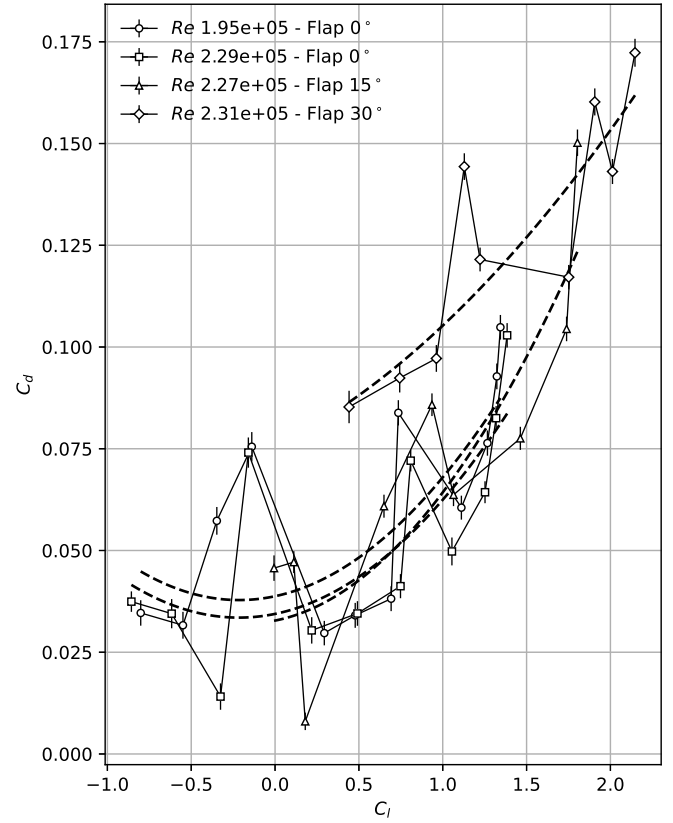
(a) For NACA 0012.



(b) For NACA 2412.



(a) For NACA 0012.



(b) For NACA 2412.

Figure 4: Drag coefficient c_d as function of the angle of attack α .

Figure 5: Parabolic fit of drag coefficient for angles of attack prior to stall.

The values of the coefficients c_{d_0} , k_1 and k_2 are shown in Tab. 6.

Table 6: Drag model coefficients got from regression.

Aerofoil	Flap	v	c_{d_0}	k_1	k_2
<i>NACA 0012</i>	0	25	$0,0159 \pm 0,001$	$0,006 \pm 0,001$	$0,016 \pm 0,002$
		35	$0,016 \pm 0,002$	$0,009 \pm 0,009$	$0,018 \pm 0,004$
<i>NACA 2412</i>	0	25	$0,042 \pm 0,009$	$0,009 \pm 0,01$	$0,01 \pm 0,02$
		35	$0,039 \pm 0,009$	$0,01 \pm 0,01$	$0,09 \pm 0,02$
	15	35	$0,03 \pm 0,06$	$0,01 \pm 0,05$	$0,02 \pm 0,03$
	30	35	$0,10 \pm 0,07$	$0,0 \pm 0,1$	$0,04 \pm 0,05$

4 Conclusion

The results confirmed the expected linear behavior of the lift coefficient for small angles of attack, particularly for the symmetric airfoil, as well as the influence of camber in increasing both the maximum lift coefficient and the lift at zero angle of attack. The cambered airfoil also exhibited a delayed decay of lift after stall compared to the symmetric case. Flap deflection further enhanced lift but produced earlier stall and increased drag, consistent with classical aerodynamic theory.

The drag measurements showed greater scatter than those of lift, reflecting the higher sensitivity of drag to experimental uncertainties and flow disturbances. Nevertheless, the pre-stall drag behavior could be reasonably represented by a parabolic model as a function of lift coefficient, especially for the symmetric airfoil.

Overall, the experimental data demonstrated good qualitative agreement with theoretical expectations, validating the proposed models within their range of applicability. The study highlights the significant role of camber and control-surface deflection in modifying airfoil performance and emphasizes the importance of experimental methods for capturing complex aerodynamic phenomena such as stall and boundary-layer separation.

References

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